

Практикум

D3:archive

Twenty questions, 62 marks. Every question the archive asks about $X \sim B(n, p)$, from May 2021 to May 2025, in the order of the six techniques rather than the order of the papers.

О практикуме

Собран долями двух тем, потому что темы «биномиальное распределение» в разметке нет: 36 баллов его вопросов лежат в `statistics.probability` (доля 0.152 — это 36 из 237) и 35 в `statistics.discrete_random_variables` (0.393 — 35 из 89). Граница между двумя темами случайна: соседние части одного вопроса и даже одна часть в двух зональных копиях ноября 2023 размечены в разные.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-d3-binomial

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

D3 archive — the binomial distribution, all of it

Twenty questions, 62 marks. Every question the archive asks about $X \sim B(n, p)$, from May 2021 to May 2025, in the order of the six techniques rather than the order of the papers.

No theory. No worked examples. Open it, answer twenty questions, close it. The theory is in the practicum, `practicum-d3-binomial.ipynb`.

§	technique	questions	marks
1	One value	4	8
2	A run of values	5	13
3	Mean and variance	3	8
4	A condition inside one distribution	3	13
5	Binomial on top of something	2	9
6	Unknown n	3	11

Answers go on one line each, to three significant figures. The checks are the same ones the practicum uses and they store nothing: each is handed the model and the event — $P(\text{flights} > 6)$ — and adds up $P(X = k)$ over the values that belong to it. When an answer is wrong because a boundary moved, the check says which boundary.

Every probability that comes from somewhere else is given. Seven of these questions take p from a normal distribution or an integral in an earlier part. Those parts are D5's and D6's; here the number is in the question.

The November 2023 paper appears once, although the archive holds it twice as TZ1 and TZ2.

Solutions are at the very bottom, deliberately far away.

```
import sys
sys.path.append('.')          # from
practicum/statistics to practicum/kit.py
import sympy as sp           # the escape
hatch: anything not in kit is in sp
from kit import *            # checks +
Bin, P(), Expect, Var, binompdf, binomcdf

language('en')               # this
notebook is in English, and so are the checks

p = symbols('p')              # the
probability of success, when it is the
unknown

print('ready; sympy', sp.__version__)
print('X ~ B(8, 0.7): P(X = 6) =',
      binompdf(8, 0.7, 6), ' P(X <= 6) =',
      binomcdf(8, 0.7, 6))
```

§ 1. One value

Four questions, 8 marks. *Exactly k:* name the model, then one button.

1.1 — May 2022 TZ1 Paper 2 Q11(b), 2 marks

The weights of chocolate muffins are normally distributed with mean 62 g and standard deviation 2.9 g, so that the probability that a randomly selected chocolate muffin weighs less than 61 g is 0.365112...

In a random selection of 12 chocolate muffins, find the probability that exactly 5 weigh less than 61 g.

1.2 — May 2023 TZ2 Paper 2 Q3(d), 2 marks

The weights of bags of rice are normally distributed, and 80% of the bags weigh between w grams and 210 grams, so that the probability that a randomly selected bag weighs less than w grams is 0.0849303...

Ten bags of rice are selected at random. Find the probability that exactly one of the bags weighs less than w grams.

```
q1_1 = ...          # P(exactly 5 of 12 muffins
under 61 g)
q1_2 = ...          # P(exactly 1 of 10 bags
under w g)

muffins = Bin(12, 0.365112)
bags = Bin(10, 0.0849303)

verify_binomial('1.1', q1_1, P(muffins == 5))
verify_binomial('1.2', q1_2, P(bags == 1))
```

1.3 — May 2024 TZ1 Paper 2 Q6(a), 2 marks

In Happyland, the weather on any given day is independent of the weather on any other day. On any day in May, the probability of rain is 0.2. May has 31 days. Find the probability that it rains on exactly 10 days in May.

1.4 — May 2023 TZ1 Paper 2 Q11(d), 2 marks

Each laboratory trial is independent, and a trial is a success with probability 0.239358... (the probability, from part (b),

that the amount of reagent used is less than 0.5 ml).

Ten trials were conducted. Find the probability that exactly three trials were successful.

```
q1_3 = ...      # P(exactly 10 rainy days)
q1_4 = ...      # P(exactly 3 successes in
10 trials)

rain = Bin(31, 0.2)
trials = Bin(10, 0.239358)

verify_binomial('1.3', q1_3, P(rain == 10))
verify_binomial('1.4', q1_4, P(trials == 3))
```

§ 2. A run of values

Five questions, 13 marks. *At most, at least, more than* — turned into $P(X \leq k)$ by you, not by the calculator.

2.1 — May 2024 TZ1 Paper 2 Q6(b), 2 marks

(Happyland, as in 1.3.) Find the probability that it rains on at least 10 days in May.

2.2 — May 2021 TZ2 Paper 2 Q10(e), 3 marks

The flight times between two cities are normally distributed, and the probability that a flight takes more than 80 minutes is 0.0711930... On a particular day, there are 64 flights scheduled between these two cities.

Find the probability that more than 6 of the flights on this particular day will have a flight time of more than 80 minutes.

```
q2_1 = ...      # P(at least 10 rainy days)
q2_2 = ...      # P(more than 6 late
flights)

rain = Bin(31, 0.2)
flights = Bin(64, 0.0711930)

verify_binomial('2.1', q2_1, P(rain >= 10))
verify_binomial('2.2', q2_2, P(flights > 6))
```

2.3 — May 2025 TZ2 Paper 2 Q10(d), 3 marks

At Adam's Apple Orchard an apple is *premium* when its weight is between 170 and 185 grams, and 62.8364...% of apples are premium. Boxes are filled with randomly chosen left-over apples; each box contains 40 apples.

Find the probability that a randomly chosen box contains at least 30 premium apples.

2.4 — May 2025 TZ3 Paper 2 Q11(a) (i), 2 marks

Amanda enters data from surveys into a database; the accuracy of any survey entered is independent of all others, and she enters 8% of the surveys

inaccurately. On a particular day Amanda enters data from 50 surveys.

Find the probability that Amanda entered at most six surveys inaccurately.

```
q2_3 = ...      # P(at least 30 premium  
apples in a box of 40)  
q2_4 = ...      # P(at most 6 inaccurate out  
of 50)  
  
box = Bin(40, 0.628364)  
surveys = Bin(50, 0.08)  
  
verify_binomial('2.3', q2_3, P(box >= 30))  
verify_binomial('2.4', q2_4, P(surveys <= 6))
```

2.5 — November 2021 Paper 2 Q3(a), 3 marks

A factory manufactures lamps. It is known that the probability that a lamp is found to be defective is 0.05. A random sample of 30 lamps is tested.

Find the probability that there is at least one defective lamp in the sample.

```
q2_5 = ...          # P(at least one defective  
lamp)  
  
lamps = Bin(30, 0.05)  
  
verify_binomial('2.5', q2_5, P(lamps >= 1))
```

§ 3. Mean and variance

Three questions, 8 marks. np , $np(1 - p)$, $a^2 \text{Var}(X)$ — and p worked back from a variance.

3.1 — May 2021 TZ2 Paper 2 Q10(d), 3 marks

(The flights of 2.2.) Find the expected number of flights that will have a flight time of more than 80 minutes.

```
q3_1 = ...          # the expected number of  
late flights  
  
flights = Bin(64, 0.0711930)  
  
verify_moment('3.1', q3_1, Expect(flights))
```

3.2 — November 2023 TZ1 Paper 2 Q6(a), 3 marks

The random variable X is such that $X \sim B(25, p)$ and $\text{Var}(X) = 5.75$. Find the possible values of p .

3.3 — November 2023 TZ1 Paper 2

Q6(b), 2 marks

The random variable Y is such that $Y = 1 - 2X$. Find $\text{Var}(Y)$.

```
q3_2 = [...]      # every possible value of p
q3_3 = ...        # Var(Y)

condition = Eq(Var(Bin(25, p)), 5.75)

verify_parameter('3.2', q3_2, condition, p)
verify_moment('3.3', q3_3, Var(1 - 2 *
Bin(25, p)), given=condition, var=p)
```

§ 4. A condition inside one distribution

Three questions, 13 marks. *Given that...*, with both events about the same count. The numerator is the overlap of two ranges.

4.1 — November 2021 Paper 2 Q3(b), 4 marks

(The lamps of 2.5.) Given that there is at least one defective lamp in the sample, find the probability that there are at most two defective lamps.

4.2 — May 2025 TZ3 Paper 2 Q11(a)(ii), 3 marks

(Amanda's 50 surveys of 2.4.) Given that at most six surveys were entered inaccurately, find the probability that exactly four surveys were entered inaccurately.

```

q4_1 = ...          # P(at most two defective |
at least one defective)
q4_2 = ...          # P(exactly four inaccurate
| at most six inaccurate)

lamps = Bin(30, 0.05)
surveys = Bin(50, 0.08)

verify_binomial('4.1', q4_1, P(lamps <= 2,
given=lamps >= 1))
verify_binomial('4.2', q4_2, P(surveys == 4,
given=surveys <= 6))

```

4.3 — November 2023 TZ1 Paper 2 Q10(c), 6 marks

The height of a wheat plant is greater than 98.1 cm with probability 0.434. The farmer measures 100 randomly selected plants. Any plant with a height greater than 98.1 cm is considered ready to harvest. Heights of plants are independent of each other.

(i) Find the probability that exactly 34 plants are ready to harvest. **(ii)** Given that fewer than 49 plants are ready to harvest, find the probability that exactly 34 plants are ready to harvest.

Write down the probability of the condition as well.

```
q4_3i = ...      # P(exactly 34 ready)
q4_3l = ...      # P(fewer than 49 ready)
q4_3 = ...      # P(exactly 34 ready | fewer
than 49 ready)

plants = Bin(100, 0.434)

verify_binomial('4.3(i)', q4_3i, P(plants ==
34))
verify_binomial('4.3(ii) condition', q4_3l,
P(plants < 49))
verify_binomial('4.3(ii)', q4_3, P(plants ==
34, given=plants < 49))
```

§ 5. Binomial on top of something

Two questions, 9 marks. Decide what one trial is, inside and outside.

5.1 — May 2025 TZ2 Paper 2 Q10(e), 2 marks

(The boxes of 2.3.) If 10 of these boxes are randomly selected, find the probability that exactly 4 boxes have at least 30 premium apples.

```
q5_1 = ...      # P(exactly 4 of 10 boxes
have at least 30 premium apples)

box = Bin(40, 0.628364)
boxes = Bin(10, P(box >= 30))

verify_binomial('5.1', q5_1, P(boxes == 4))
```

5.2 — May 2022 TZ2 Paper 2 Q8, 7 marks

Rachel and Sophia are competing in a javelin-throwing competition. The distances thrown by Rachel are normally distributed with mean 56.5 m and standard deviation 3 m, and those

thrown by Sophia with mean 57.5 m and standard deviation 1.8 m; so for one throw

$$P(\text{Rachel} \geq 60) = 0.1216725\dots, \quad P(\text{Sophia} \geq 60) \\ = 0.0824333\dots$$

In the first round of competition, each competitor must have five throws. To qualify for the next round of competition, a competitor must record at least one throw of 60 metres or greater in the first round.

Find the probability that only one of Rachel or Sophia qualifies for the next round of competition.

```
q5_2r = ...      # P(Rachel qualifies)
q5_2s = ...      # P(Sophia qualifies)
q5_2 = ...      # P(only one of them
qualifies)

rachel = Bin(5, 0.1216725, 'R')
sophia = Bin(5, 0.0824333, 'S')

verify_binomial('5.2 Rachel', q5_2r, P(rachel
>= 1))
verify_binomial('5.2 Sophia', q5_2s, P(sophia
>= 1))
verify_binomial('5.2', q5_2, P((rachel >= 1)
^ (sophia >= 1)))
```

§ 6. Unknown n

Three questions, 11 marks.

Logarithms where they work, a table where they do not, and a whole number at the end either way.

6.1 — May 2024 TZ2 Paper 2 Q5, 5 marks

Consider a random variable X such that $X \sim B(n, 0.25)$. Determine the least value of n such that $P(X \geq 1) > 0.99$.

6.2 — May 2023 TZ1 Paper 2 Q11(c), 3 marks

(The laboratory trials of 1.4, each a success with probability 0.239358...)

Determine the least number of trials required to be 99% sure of at least one success.

```

q6_1 = ...          # the least n
q6_2 = ...          # the least number of trials

verify_trials('6.1', q6_1, lambda n: P(Bin(n,
0.25) >= 1), holds=lambda value: value >
0.99)
verify_trials('6.2', q6_2, lambda n: P(Bin(n,
0.239358) >= 1),
              holds=lambda value: value >=
0.99)

```

6.3 — May 2025 TZ3 Paper 2 Q11(b), 3 marks

On a different day Amanda (*of 2.4*) enters data from n surveys, each inaccurate with probability 0.08. On this day, the probability that at most six surveys were entered inaccurately is approximately 0.367.

Find the value of n .

```

q6_3 = ...          # n

verify_trials('6.3', q6_3, lambda n: P(Bin(n,
0.08) <= 6), near=0.367)

```



Solutions

1.1 $M \sim B(12, 0.365112\dots)$, and $P(M = 5) = 0.213666\dots = \boxed{0.214}$. The M1 is for writing the model.

1.2 $L \sim B(10, 0.0849303\dots)$, and $P(L = 1) = 0.382076\dots = \boxed{0.382}$.

1.3 $X \sim B(31, 0.2)$: $P(X = 10) = \binom{31}{10}(0.2)^{10}(0.8)^{21} = 0.0418894\dots = \boxed{0.0419}$.
Without working, 0.042 scores (M1)A0.

1.4 $Y \sim B(10, 0.239358\dots)$, and $P(Y = 3) = 0.242430\dots = \boxed{0.242}$.

2.1 *At least 10* is everything but $0, \dots, 9$:
 $P(X \geq 10) = 1 - P(X \leq 9) = 1 - 0.925400\dots = \boxed{0.0746}$. $1 - P(X \leq 10) = 0.0327$ is *more than 10*.

2.2 *More than 6* starts at 7: $P(L > 6) = 1 - P(L \leq 6) = 1 - 0.830880\dots = \boxed{0.169}$.

2.3 $A \sim B(40, 0.628364\dots)$: $P(A \geq 30) = 1 - P(A \leq 29) = 0.073861\dots = \boxed{0.0739}$.

2.4 $E \sim B(50, 0.08)$: $P(E \leq 6) = 0.898128 \dots = \boxed{0.898}$.

2.5 $D \sim B(30, 0.05)$: $P(D \geq 1) = 1 - 0.95^{30} = 0.785361 \dots = \boxed{0.785}$.

3.1 $E(L) = np = 64 \times 0.0711930 \dots = 4.55635 \dots = \boxed{4.56}$ flights. An expected number is not rounded to a whole one.

3.2 $25p(1-p) = 5.75$, so $25p^2 - 25p + 5.75 = 0$ and $p = \frac{1}{2} \pm \frac{\sqrt{2}}{10}$: $\boxed{p = 0.641 \text{ or } 0.359}$. A1A1 — both are needed, and they add to one because p and $1-p$ give the same variance.

3.3 $\text{Var}(1 - 2X) = (-2)^2 \text{Var}(X) = 4 \times 5.75 = \boxed{23}$.

4.1 The values in both *at least one* and *at most two* are 1 and 2:

$$\begin{aligned}
 P(D \leq 2 \mid D \geq 1) &= \frac{P(D = 1) + P(D = 2)}{P(D \geq 1)} \\
 &= \frac{0.597540 \dots}{0.785361 \dots} \\
 &= \boxed{0.761}
 \end{aligned}$$

Recognition of the conditional must be shown in context, not as a bare $P(A \mid B)$.

4.2 *Exactly four lies inside at most six:*

$$P(E = 4 \mid E \leq 6) = \frac{0.203654 \dots}{0.898128 \dots} = \boxed{0.227}$$

4.3 (i) $H \sim B(100, 0.434)$: $P(H = 34) = 0.0133198 \dots = \boxed{0.0133}$.

(ii) *Fewer than 49 is* $H \leq 48$, and $P(H \leq 48) = \boxed{0.848}$. Then

$$\begin{aligned}
 P(H = 34 \mid H < 49) &= \frac{0.0133198 \dots}{0.848218 \dots} \\
 &= \boxed{0.0157}
 \end{aligned}$$

Using $P(H \leq 49) = 0.890474 \dots$ gives 0.0149581, and the markscheme names it: (A0)(M1)(A1)A0.

5.1 A trial is now a box: $Y \sim B(10, 0.073861 \dots)$ and $P(Y = 4) =$

$$0.003944\dots = \boxed{0.00394}.$$

5.2 Each competitor qualifies with at least one long throw in five:

$$\begin{aligned} P(N_R \geq 1) &= 1 - (1 - 0.1216725\dots)^5 \\ &= \boxed{0.477}, \quad P(N_S \geq 1) \\ &= 1 - (1 - 0.0824333\dots)^5 \\ &= \boxed{0.350} \end{aligned}$$

Only one of them:

$$\begin{aligned} &0.477264\dots \times 0.650412\dots \\ &+ 0.349588\dots \times 0.522736\dots = \boxed{0.493} \end{aligned}$$

At least one would be 0.660 — the case where both qualify is not *only one*.

6.1 $1 - 0.75^n > 0.99$, so $0.75^n < 0.01$ and $n > \frac{\ln 0.01}{\ln 0.75} = 16.0078\dots$: $\boxed{n = 17}$. By table, $n = 16$ gives 0.989977... and $n = 17$ gives 0.992483...

6.2 $1 - (0.760641\dots)^n \geq 0.99$ gives $n \geq 16.8321\dots$: $\boxed{17}$ trials. By table, $n = 16$ gives 0.987443... and $n = 17$ gives 0.990448...

6.3 $P(E \leq 6)$ for $E \sim B(n, 0.08)$ falls as n grows: $n = 93$ gives 0.378, $n = 94$ gives 0.367, $n = 95$ gives 0.356. $\boxed{n = 94}$. The markscheme's M1 is for any correct value of $P(E \leq 6)$ with $n \neq 50$ — the table is the method.